Tableau Public Tutorial

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Tableau Public Tutorial: Analyzing Airbnb Data

Introduction

This tutorial will guide you through using Tableau Public to analyze Airbnb listing data, starting from the very basics. Each step includes specific instructions about where to find buttons, tools, and features in the interface.

##Background

Airbnb launched in 2009 by two friends who rented out their San Francisco apartment for a local convention, and it is now one of the largest hotel/rental businesses in the world. In Q3 of 2022, Airbnb has had nearly 100 million nights and experiences booked (Airbnb, 2023). Airbnb uses a commission-based platform that allows hosts to rent out a room or the entire dwelling of their home. The type of rental can vary from a

single room in Austin, Texas to an entire penthouse in Manhattan, New York. One feature of the platform is the historical records of reviews by past customers. These records help potential customers gauge the popularity of the property. It also helps build on the property information that is supplied by the renter (Aznar et al, 2016). According to a survey by Goldman Sachs, a person's likelihood to prefer a traditional hotel is cut in half if they have stayed in a shared space rental property within the last five years (Clever Real Estate, 2021). Airbnb has significantly disrupted the traditional hotel industry. This new sharing model goes against the traditional hotel model that is characterized by having an entry barrier, investments, and operating at a fixed cost (Aznar et al, 2016). Because of the lower entry barrier, Airbnb and other shared rental services continue to grow. According to a study of 1,000 people by Clever in 2019, 82% said they would consider renting a portion of their place for extra income (Clever Real Estate, 2021). According to a survey conducted by Earnest, on average, Airbnb hosts make over $500 per month ($6,000 annually) (Morris, 2021). As part of the rental process, Airbnb charges a flat 3% service fee to each host. This fee covers Airbnb services such as customer service, marketing, and insurance (Airbnb, 2020). As part of this case study, you will take the role of a data analyst. Your job is to look at a data set of Airbnb rentals from New York City. Using this data, identify the profitability of hosts in the city. Along with identifying profits, use the data to identify which areas of the city have grown in popularity and which areas have not. This assignment uses a free Airbnb data set published on the Inside Airbnb website. The goal of this assignment is to provide a familiarity with common Tableau features including maps, graph types, filters, mark types, calculated fields, formatting, and aggregation. A walkthrough solution of the prompt is provided.

Prerequisites

- Download and install Tableau Public from public.tableau.com
- Download the provided Airbnb_listing.csv file
- Create a free Tableau Public account

Part 1: Getting Started

1.1 Opening Tableau Public & Loading Data

1. Launch Tableau Public by double-clicking its icon on your desktop or finding it in your applications
2. When the application opens, you'll see the start page. Look for the large orange button labeled "Create a Visualization" in the center of the screen
3. After clicking, you'll see the "Connect to Data" screen. On the left side of this screen, under "Connect", find and click "Text file"
4. A file browser window will open. Navigate to where you saved Airbnb_listing.csv
5. Select the file and click "Open"
6. You'll now see the "Data Source" page where you can preview your data

1.2 Understanding the Data Source Page

The Data Source page is where you prepare your data. Here's what you're looking at: 1. Top left: The name of your data source (Airbnb_listing.csv) 2. Left panel: Lists all available data connections 3. Center: Preview of your data in a grid format 4. Bottom: Tabs for "Connection" and "Data Source" 5. Top right: "Go to Worksheet" button (orange)

1.2.1 Getting to know your data

The data used in this study is sourced from the public information on the Airbnb site. Information from this site is collected and aggregated by Inside Airbnb to help facilitate public discussion. A definition of the data types is provided below.

Field Name	Field Type	Description	Example
ID	Number	Unique identifier for the listing	5136
Name	String	Name of the listing	Midtown Castle
HostId	Number	Airbnb's unique identifier for the listing	5162530
Host Name	String	Name of the host	Steve
Host Since	Date	The date the host/user was created	6/26/08
Neighborhood	String	Neighborhood where listing is located	Williamsburg
Neighborhood Group	String	Borough where listing is located	Brooklyn
Latitude & Longitude	Number	Uses the World Geodetic System (WGS84) projection for latitude and longitude	40.764570,-73.955120
Room Type	String	Host's description of the room	Private room
Accommodates	Number	The maximum capacity of the listing	4
Bathrooms	String	The number of bathrooms in the listing	1 bath
Bedrooms	Number	The number of bedrooms	3
Price	Number	Daily price in local currency ($)	145
Beds	Number	The number of bed(s)	1
Number of Reviews	Number	The number of reviews the listing has. Reviews are left after a rental	171
Last Review	Date	The date of the last/newest review	11/21/2022

1.3 Navigating to the Workspace

1. Look for the orange "Sheet 1" tab at the bottom of the screen
2. Click this tab to enter the main Tableau workspace

1.4 Understanding the Main Workspace

The Tableau workspace has several key areas. Let's explore each one in detail:

Left Sidebar - Data Pane Located on the far left of your screen, the Data Pane contains: 1. **Data Source section** (top) - Shows your connected data (Airbnb_listing.csv) - Click the dropdown arrow to see options

2. **Fields section** (below Data Source)
   - **Dimensions** (blue fields)
     - These are categorical data like neighborhood, room_type
     - Usually text or dates

- **Measures** (green fields)
  - These are numerical data like price, number_of_reviews
  - Can be aggregated (summed, averaged, etc.)

Top Bar - Shelves At the top of the workspace, you'll find: 1. **Pages Shelf** (leftmost) - Used for creating animations or step-through views 2. **Filters Shelf** (next to Pages) - Where you add filters to limit your data 3. **Marks Shelf** (next) - Controls how your data points look 4. **Columns Shelf** (top) - Fields dropped here create vertical divisions 5. **Rows Shelf** (second from top) - Fields dropped here create horizontal divisions

Center - Canvas The main workspace where your visualization appears: 1. **Grid** - Shows when empty - Has "Drop field here" indicators 2. **Visualization Area** - Where your charts/graphs appear - Updates in real-time as you make changes

Right Side - Cards and Controls From top to bottom: 1. **Show Me** (toggle button in top right) - Click to show/hide - Displays all possible visualization types - Highlights which ones are suitable for your selected fields

2. **Marks Card** (below Show Me)
   - **Color**: Controls the colors in your visualization
   - **Size**: Adjusts the size of marks
   - **Label**: Adds data labels
   - **Detail**: Adds more information without changing the view
   - **Tooltip**: Customizes hover information

Bottom Bar - Navigation At the bottom of the screen: 1. **Sheet Tabs** - Shows all your worksheets - Plus sign (+) to add new sheets 2. **Dashboard Tab** - For combining multiple visualizations 3. **Story Tab** - For creating guided presentations

1.5 Basic Interface Interactions

Let's practice some basic interactions:

1. **Moving Fields**
   - Click and drag any field from the Data Pane
   - Notice how the visualization area shows blue or green guides
   - Blue means the field will be used as a dimension
   - Green means it will be used as a measure
2. **Changing Mark Types**
   - Find the Marks card dropdown (currently says "Automatic")
   - Click to see all mark types (bar, line, circle, etc.)
   - Try selecting different ones to see what changes
3. **Show Me Practice**
   - Click "Show Me" in the top right
   - Notice how some icons are grayed out
   - Select multiple fields in your Data Pane
   - Watch how available chart types change
4. **Undo/Redo**

- Look for the undo (←) and redo (→) arrows in the top toolbar
- Try making a change and undoing it
- These buttons will be your best friends! Take a moment to explore these areas and familiarize yourself with the layout.

Part 2: Creating Your First Visualization

2.1 Bar Chart of Top Rental Areas

1. In the Data Pane, find "neighbourhood"
2. Drag "neighbourhood" to Rows
3. Drag "id" to Columns
4. In the Marks card:
   - Change the mark type to "Bar"
   - Drop "neighbourhood" onto Color
5. Right-click on the id axis and select "Measure" → "Count"
6. Sort the bars by clicking the sort button in the toolbar
7. Filter to top 10 neighborhoods:
   - Right-click "neighbourhood" in Rows
   - Select "Filter"
   - Choose "Top" tab
   - Select "By Field"
   - Set to "Top 10" by "Count of id"
8. Rename the sheet by double-clicking on the table name above your chart. Call your table something that makes sense. "Top 10 AirBnB neighborhoods in New York City." or something similar.

Part 3: Working with Filters

3.1 Adding Interactive Filters

1. Create a new worksheet
2. Drag "room_type" to Filters
3. Right-click the filter and select "Show Filter"
4. Repeat with "host_since" (you'll need to create a date filter)
5. Configure the date filter:
   - Select "Years"
   - Choose "All" to show all available years

3.2 Applying Filters to Multiple Sheets

1. Right-click your filter
2. Select "Apply to Worksheets"
3. Choose "All Using This Data Source"

Part 4: Spatial Analysis

4.1 Creating a Map Visualization

1. Create a new worksheet

2. Double-click on “Latitude” and “Longitude”
3. Tableau will automatically create a map
4. Drag “price” directly to Size in the Marks card
    - The values are already in numeric format, so no cleaning is needed
5. Drag “neighbourhood” to Color
6. Add tooltips with listing details:
    - Click “Tooltip” in the Marks card
    - Add fields: name, room_type, price, neighborhood, neighborhood group

Part 5: Creating Calculated Fields

5.1 Calculate Host Revenue

1. Right-click in the Data Pane
2. Select “Create Calculated Field”
3. Name it “Estimated Revenue”
4. Enter the formula:

```
[price] * [number_of_reviews] * 0.97
```

5. Click OK

5.2 Creating Revenue Analysis

1. Create a new worksheet
2. Drag your new “Estimated Revenue” field to Columns
3. Drag “host_name” to Rows
4. Sort by revenue to see top-earning hosts

Part 6: Building a Dashboard

6.1 Creating the Dashboard

1. Click the “New Dashboard” icon
2. Set size to “Automatic” for web publishing
3. Drag your sheets from the left sidebar:
    - Map visualization
    - Revenue analysis
    - Top neighborhoods
4. Arrange them in a logical layout
5. Add a dashboard title
6. Add interactive filters that apply to all visualizations

6.2 Adding Interactivity

1. Click on any visualization
2. Select “Use as Filter”
3. Test the interaction by clicking different areas

Part 7: Creating a Story

7.1 Building Your Story

1. Click "New Story" in the bottom tabs
2. Add a story title: "Airbnb Market Analysis"
3. Create story points:
    - Market Overview (dashboard)
    - Neighborhood Analysis (top areas)
    - Pricing Patterns (map)
    - Host Performance (revenue)
4. Add captions to explain insights

Part 8: Publishing

8.1 Publishing to Tableau Public

1. Click "File" → "Save to Tableau Public"
2. Sign in to your Tableau Public account
3. Enter a workbook name
4. Add a description
5. Set visibility options
6. Click "Save"

8.2 Sharing Your Work

1. Once published, copy the share link
2. Set visibility preferences:
    - Public (anyone can see)
    - Hidden (only via direct link)
3. Embed options are available for web integration

Best Practices

- Always clean your data before analysis
- Use clear titles and labels
- Include explanatory notes in tooltips
- Test your filters and interactions
- Save your work frequently
- Consider color-blind friendly palettes

Troubleshooting Common Issues

- Data connection errors: Verify CSV format
- Map not showing: Check latitude/longitude fields
- Calculation errors: Verify formula syntax
- Publishing issues: Check internet connection

Next Steps

- Explore other visualization types
- Learn about advanced calculations
- Practice with different datasets
- Join the Tableau community
- Share your visualizations

Remember to save your work regularly and experiment with different visualization types to find the most effective way to communicate your insights.

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

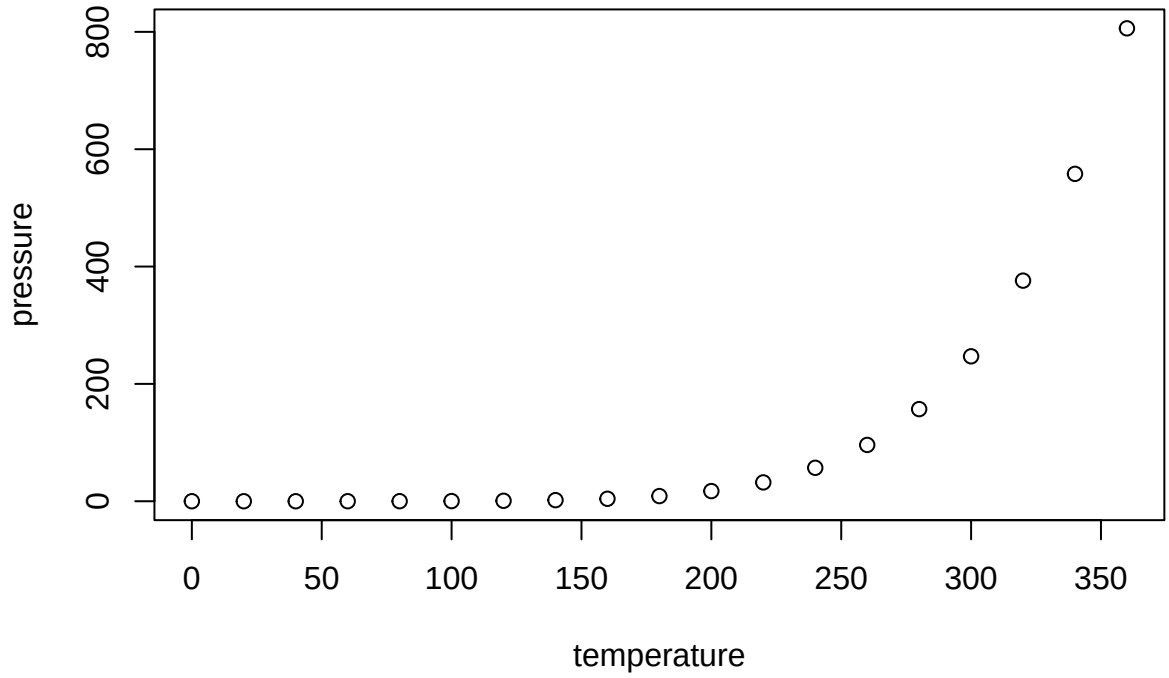
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```r
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.